

Audio Deep Learning

2025/2026, Summer Trimester

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Department	Computer Science Department
Study program	Artificial Intelligence
ECTS credits	4
Class hours	52 hours, 26 lecture hours and 26 practice hours
Course language	English
Course format	Offline
Form of study	Full-time studies

Overview

Prerequisites

This course assumes that students already possess a solid foundation in mathematics, statistics, data manipulation, machine learning, and deep learning. The prerequisites are specified in terms of dependent courses, hard skills, and soft skills.

Dependent courses.

Must-Have (Essential). Students must have successfully completed the following courses (or demonstrate equivalent knowledge):

- *Data Manipulation Essentials* – familiarity with working on structured and unstructured data, data cleaning, transformations, and visualization.
- *Applied Linear Algebra* – concepts of vector spaces, matrix operations, eigenvalues/eigenvectors, orthogonality and projections.
- *Probability Theory and Statistics* – probability distributions, expectation/variance, conditional probability, Bayes' theorem.
- *Optimization Techniques* – gradient descent, constrained optimization, and regularization.
- *Machine Learning* – supervised and unsupervised learning, model evaluation and basic ML pipelines.
- *Deep Learning* – the main building blocks and architectures of neural networks, the concept of the training loop, various specialized optimizers, and regularization techniques.
- *Computer Vision* - dedicated architectures, building blocks, optimization and training strategies for Computer Vision tasks. Together with overall understanding of different Computer Vision tasks, like Classification, Segmentation, Detection, Self-Supervised Learning
- *Natural Language Processing* - overview of key architectures, building blocks, optimization techniques, and training strategies relevant to sequence modeling tasks. Particular emphasis is placed on language modeling approaches and large language model (LLM) architectures, including their training and inference strategies. The section also covers fundamental text processing and representation methods, which play a critical role in automatic speech recognition systems.

Hard skills.

- *Programming:* Proficiency in Python, including NumPy, pandas, and matplotlib.
- *Math Foundations:* Linear algebra: dot product, matrix multiplication, eigen decomposition.
- *Calculus:* differentiation, gradients, partial derivatives.
- *Probability:* basic distributions, expectation, variance, Bayes' theorem.
- *Familiarity with gradient-based optimization* (e.g., gradient descent).
- *Data Handling:* Loading, preprocessing, and transforming datasets for ML tasks.
- *Familiarity with major deep learning architectures:* Feedforward Networks, basic CNNs (e.g., ResNet), Recurrent Neural Networks (RNN, GRU, LSTM), and Transformers (BERT, GPT).

Soft skills.

- *Self-Learning Ability:* Comfort with reading technical documentation and papers.
- *Analytical Thinking:* Ability to break down problems and choose appropriate solution strategies.
- *Time Management:* Capable of balancing coursework, projects, and self-study.
- *Collaboration:* Experience working in small project teams.
- *Communication:* Ability to present technical findings to mixed-knowledge audiences.
- *Problem Formulation:* Skill in framing real-world problems into ML tasks.

Background and course rationale

Audio Deep Learning is a core field of artificial intelligence that focuses on enabling machines to perceive, interpret, and reason about sound. Modern audio systems are largely powered by deep learning, which has led to major breakthroughs in speech recognition, speaker analysis, audio classification, sound event detection, diarization, and speech synthesis. Together with related areas of signal processing such as processing of medical signals (EEG), radio frequencies, gravitational waves, and others

This course provides a solid foundation in the fundamental concepts, architectures, and practical methods of deep learning for audio processing. It connects classical signal processing techniques with modern data-driven approaches, bridging theoretical machine learning principles and hands-on implementation of state-of-the-art audio models.

The course addresses questions such as:

- *How can classical audio signal processing techniques be applied to real-world tasks in speech and sound analysis?*
- *How can neural network architectures be used across a wide range of audio problems, including classification, diarization, automatic speech recognition, and text-to-speech synthesis?*
- *How can self-supervised learning methods be leveraged to learn robust audio representations and improve performance under limited labeled data?*
- *How should audio processing systems be designed for different application domains, such as speech technologies, multimedia analysis, and real-world industrial or medical scenarios?*

The course covers fundamental audio representations, including waveform and spectral features, along with classical preprocessing techniques such as filtering and normalization. It introduces modern deep learning approaches for audio, including representation learning, classification models, and sequence modeling methods used in automatic speech recognition (ASR), such as CTC-based architectures and Transformer models (e.g., Whisper). The course also explores speaker diarization and advanced generative approaches for speech synthesis, including sequence-to-sequence models, language modeling approaches, and diffusion-based text-to-speech systems. Additionally, the course emphasizes self-supervised learning techniques for audio representation learning, focusing on improving robustness and performance in data-scarce settings.

Course aims

The primary goals of this course are to:

- *Introduce the fundamental theory and methods of Digital Signal Processing (DSP) for analyzing analog signals in digital systems, including the application of digital filters and other signal transformations.*
- *Present both fundamental and state-of-the-art neural network architectures and deep learning methods for signal-related tasks, while developing the ability to design and combine novel architectural building blocks.*
- *Cover a broad range of audio-related tasks, including signal classification, speaker diarization, automatic speech recognition, and speech generation, with hands-on experience on real-world problems.*
- *Explore how popular architectures and methods from natural language processing and computer vision can be adapted and applied to audio-related tasks.*

Learning outcomes

By the end of the course, students will be able to:

- Explain the principles, strengths, and limitations of classical digital signal processing methods and identify appropriate scenarios for their application.
- Apply deep learning architectures to audio-related tasks, including signal classification, speaker diarization, automatic speech recognition, and speech synthesis.
- Design, modify, and combine neural network components to build custom models for audio and time-series data.
- Select and adapt signal processing and deep learning methods for different application domains, such as speech technologies, healthcare, and real-world industrial systems.
- Utilize self-supervised learning techniques to learn robust audio representations, particularly in settings with limited labeled data.
- Analyze model performance and reason about trade-offs between accuracy, computational efficiency, latency, and robustness in audio systems.
- Develop and evaluate end-to-end audio processing pipelines using real-world datasets.

Methods of Teaching and Assessment

The course uses a blended learning approach combining theoretical lectures, practical coding exercises, independent study, and research-oriented activities. The aim is to ensure that students not only understand the underlying concepts but also gain the ability to implement, experiment with, and critically evaluate deep learning solutions.

Methods:

- Lectures – Weekly sessions introducing theoretical concepts, mathematical foundations, and real-world case studies. Delivered with live coding demonstrations where applicable.
- Practical Labs – Hands-on coding sessions in which students implement algorithms and architectures both from scratch and using PyTorch/Lightning.
- Project Work – A semester-long project allowing students to apply deep learning methods to a real-world dataset or problem of their choice, culminating in a final report and/or presentation.
- Kaggle Competitions – Participation in curated challenges to apply learned techniques under realistic, competitive conditions.
- Independent Study – Structured self-learning tasks, including preparatory readings and follow-up exercises.

Course Faculty



Volodymyr Sydorskyi

Research Engineer | R&D Lead & AI Startup Co-Founder | Kaggle Competition Grandmaster | Educator & PhD Candidate

I'm Volodymyr, and my main passion lies in pure Deep Learning—free from deployment headaches and other boring stuff :)

I've led the development of some of the best-performing voice conversion and text-to-speech systems. Building on that experience, together with colleagues, I developed an acoustic detection system for identifying airborne threats (missiles, drones, jets, helicopters). Both systems are successfully operating today—the first has been

used in Hollywood productions like *The Mandalorian* and AAA games such as *Cyberpunk*, while the second helps protect the Ukrainian sky every day.

I'm also deeply engaged in competitive machine learning. You may have seen my name in the Kaggle BirdCLEF competition or other audio and computer vision challenges. Alongside in a team of ML practitioners, I'm a regular participant in local hackathons. After a long competitive journey, I achieved the Kaggle Competition Grandmaster title and currently rank among the top competitors worldwide.

Another area close to my heart is academia. I'm pursuing a PhD in System Analysis at the Institute for Applied System Analysis (KPI), focusing on medical imaging and building a fully end-to-end system for cancer localization and classification. This research aligns with my work as a Senior Computer Vision Engineer at Better Medicine, where we develop cancer detection systems. I've also co-designed and taught university-level courses in NLP and Audio ML at KPI and UCU—materials for which are available on my [GitHub](#) and [YouTube](#). Conducting research and publishing papers, especially in collaboration with talented students and colleagues, is one of my favorite pursuits.

As for my philosophy, don't follow trends blindly and don't waste time on uninspiring work. Instead, chase ambitious, crazy, and inspiring ideas. Some will fail, but a few will shape new trends or become groundbreaking products and scientific breakthroughs.



Danil Brovko

Deep Learning Research Engineer | Scalable AI Systems | M.Sc. in AI

You can find my passion by calculating the dot product between the diverse vectors of my life: from detecting landmines using hyperspectral imagery to synthesizing AI voices for Hollywood.

In the corporate world, I have navigated a broad spectrum of challenges, building high-load distributed systems, creating AI assistants, and cloning voices for Oscar-winning films and AAA games.

I hold both a B.Sc. and M.Sc. in AI from IASA at Igor Sikorsky Kyiv Polytechnic Institute, with plans to continue my academic journey with a PhD. My research has evolved alongside my interests. My Bachelor's resulted in a Springer chapter on PolliWeather – a system to track and predict pollen hazards in Kyiv, while my Master's work led to 1st place at an international AI competition and an IEEE conference paper on SpectralCA – a pre-trained custom architecture for detecting landmines using hyperspectral data.

I believe that true understanding comes from hands-on practice, when you start asking and exploring: "Why exactly is it done this way?" Dream big, question the default, and make something great.

Let's finish with a quote by Sun Tzu: "Victorious warriors win first and then go to war, while defeated warriors go to war first and then seek to win". Better be the former on your lab defence!

Course Structure

Course Duration:

- Length: 11 weeks (1 trimester)
- Credits: 4 ECTS

- *Workload: 26 hours lectures, 26 hours practical labs, 20–30 hours independent study and assignments, 20–30 hours course project and additional activities*

Lectures

- *Format: In-person live sessions with interactive Q&A and occasional live coding.*
- *Content: Theoretical concepts, mathematical foundations, real-world case studies, and architecture overviews.*
- *Materials: Slides and code notebooks provided via the course repository after each lecture.*
- *Attendance: Strongly recommended.*
- *Prerequisites: Some lectures require reviewing provided readings or tutorials beforehand*

Module 1 – Intro in Audio. Basic Audio Processing

Lecture 1 – Intro in Audio and Audio ML. Digital wave representation. Spectral audio representation (Volodymyr): Overview of analog and digital audio signals, including waveform representation and fundamental spectral analysis techniques. Discussion of key challenges in digital signal processing for speech and audio data.

Lecture 2 – Filtering, Audio Normalization (Volodymyr): Introduction to commonly used digital filters and practical techniques for audio normalization and preprocessing in real-world applications.

Lecture 3 – Self Supervised Audio Representations (Volodymyr): Introduction to self-supervised learning in the audio domain, including core concepts and an overview of major self-supervised models for learning robust audio representations.

Module 2 – Audio Classification and Diarization

Lecture 1 – Audio Classification model (Volodymyr): Introduction to audio classification tasks and problem formulations. Overview of common model architectures, including convolutional and transformer-based approaches. Discussion of training strategies, data augmentation techniques, and evaluation metrics for real-world audio classification problems.

Lecture 2 – Speaker diarization (Volodymyr): Introduction to the speaker diarization problem (“who spoke when”) and its key components. Overview of segmentation, speaker embedding extraction, and clustering methods. Discussion of modern deep learning approaches for diarization, evaluation protocols, and practical challenges such as overlapping speech and domain variability.

Module 3 – Automatic Speech Recognition

Lecture 1 –Introduction to Generative Models (Volodymyr): Comprehensive overview of generative modeling approaches for sequential and audio data. The lecture covers a wide range of modern generative paradigms, including Variational Autoencoders (VAE), Generative Adversarial Networks (GANs), autoregressive models, diffusion models, and flow matching methods.

Lecture 2 –Introduction to ASR (Danil): Overview of automatic speech recognition systems, including problem formulation and core components. Introduction to modern end-to-end ASR systems. Discussion of acoustic modeling, language modeling, decoding strategies, evaluation metrics (e.g., WER), and key challenges in real-world applications.

Lecture 3 – Deeper ASR overview. CTC loss and Encoder architecture for ASR (Danil): In-depth analysis of modern ASR architectures with a focus on encoder-based models. Introduction to Connectionist Temporal Classification (CTC) loss, alignment-free training, and decoding methods. Overview of encoder architectures (CNNs, RNNs, Transformers).

Lecture 4 - Language Modeling-based models. Whisper(Danil): Introduction to language modeling approaches for speech recognition, including sequence-to-sequence Transformer architectures. Detailed overview of the Whisper model, its training paradigm, and practical usage. Discussion of multilingual ASR, robustness to noise, and integration of language modeling into end-to-end speech systems.

Module 4 – Speech Generation

Lecture 1 – Sequence-2-Sequence Text To Speech models (Danil): Introduction to sequence-to-sequence approaches for text-to-speech synthesis. Overview of encoder–decoder architectures, attention mechanisms, and alignment learning between text and audio. Discussion of classical and modern TTS systems (e.g., Tacotron-like models)

Lecture 2 – Language Modeling-based models TTS (Danil): Exploration of language modeling approaches to speech generation, where audio is treated as a sequence modeling problem. Overview of autoregressive and Transformer-based models for TTS, discrete audio tokenization, and large-scale pretraining.

Lecture 3 – Diffusion Approach for TTS (Danil): Introduction to diffusion-based generative models for speech synthesis. Explanation of forward and reverse diffusion processes, noise modeling, and training objectives. Discussion of advantages over autoregressive methods, including improved quality and diversity, as well as computational trade-offs.

Lecture 4 - Speech Conversion (Volodymyr): Overview of voice and speech conversion tasks, including speaker identity transformation, style transfer, and cross-lingual voice adaptation. Discussion of representation learning, disentanglement of content and speaker characteristics, and modern deep learning approaches for high-quality and controllable speech conversion.

Practical Labs

- *Format: Hands-on coding in Jupyter or PyTorch Lightning, guided by instructors/lecturers, presentations and defenses of lab assignments, paper review and project presentations.*
- *Focus: Implementing neural network architectures both from scratch and using modern frameworks.*

Lab Assignments:

- *Format: Individual or group (max 3 students) with offline defenses. Up to 3 attempts to defense and rework the lab before the deadline are possible*
- *Grading: 4 graded lab assignments (17 points each)*
- *Deadlines: Assignment submissions due 2 weeks (hard deadline) after the corresponding lab session*

Lab 1 – Ukrainian Speech Emotion Recognition using Self-Supervised Models and DSP Processed Audio (17 points):

Students will preprocess raw Ukrainian speech audio using digital signal processing techniques, extract or utilize self-supervised audio representations, and build an emotion recognition system. The lab includes data preparation, feature extraction or model fine-tuning/adaptation, inference, and evaluation using appropriate classification metrics such as accuracy, macro F1-score, and per-class performance.

Lab 2 – Solve Audio Classification task (17 points)

Students will develop an end-to-end audio classification pipeline for a real-world task. The lab includes feature preparation, model training, data augmentation, validation, and performance analysis using standard classification metrics.

Lab 3 – Build Speech Recognition System(17 points)

Students will implement and evaluate an automatic speech recognition system using modern deep learning approaches. The lab includes dataset preparation, model training or fine-tuning, decoding, and assessment of recognition quality with speech-specific evaluation metrics.

Lab 4 – Build Text To Speech system(17 points)

Students will build a text-to-speech synthesis pipeline based on modern neural approaches. The lab includes text and audio preprocessing, model training or adaptation, waveform generation, and qualitative and quantitative evaluation of synthesized speech.

Course Project

- *Type: Individual or group (max 5 students).*
- *Goal: Solve a real-world audio- or signal-related problem using methods from the course.*
- *Topics: either suggested by students or could be a project based on the kaggle competition or research project based on “future work”-like sections of any suitable scientific paper*
- *Grading: 25 points (20 for implementation and results, 5 for presentation/report). When presenting a project, it is important to disclose who on the team did what exactly for this project.*

Preliminaries test

- *Type: Individual written assignment to evaluate the preliminary knowledge of the student.*
- *Grading: 7 points.*
- *Timeline: During the 1st practice session*

Additional Activities

You can get up to +30 bonus points from these activities:

- *Kaggle Competitions: Bonus points for the private leaderboard result: Bronze: +5 points, Silver: +15 points, Gold: +30 points*
- *Research Contributions: up to +20 bonus points for accepted conference/workshop papers. The specific score 1-20 will be determined after agreement with the course faculty, depending on the level of the journal or conference*

Grading Policy

The maximum total points in the course is 100. However, students can earn up to 130 points by completing bonus activities, which allows them to exceed 100% and improve their grade even if they lose points elsewhere. Any points above 100 will be considered extra credit but will not exceed the maximum recorded grade of 100%.

Activity	Points
Preliminaries test	7
Lab 1	17
Lab 2	17
Lab 3	17
Lab 4	17
Course Project	25
Total	100
Bonus Activities	up to 30

Attendance Policy

- Lecture attendance is recommended but not graded.
- Lab Assignment defenses attendance is mandatory – absence without prior notice and approval will result in a 0 for that lab defense unless made up within the allowed timeframe.

Preliminaries test (7 points)

- Format: Written assignment with described points per each question/task during test
- Grading: 7 points.
- Timeline: During the 1st practice session

Lab Assignments (4x17 points) Defense Policy:

- Format: Individual or group (max 3 students) with offline defense
- Grading: 4 graded lab assignments, each worth 17 points
- Deadlines: Assignment submissions are due 2 weeks after the corresponding lab session (hard deadline; no late submissions accepted unless pre-approved for exceptional circumstances)
- The defense session includes: Presentation of the solution, Live code review, Theoretical Q&A related to the lab topic

- Every student in the group will be asked individual questions on both the code and the theory to ensure understanding.
- A student who cannot answer questions on their group's code or theory may receive a reduced grade for that lab.
- On Practice small written test will be handled with 1-2 practice tasks

Criterion	Points
Overall correctness and completeness of the submitted code for the lab assignment, overall quality (metrics) of the proposed solution	5
Practical question about the submitted code during the defense	7
Theoretical question during defense	2
Practice written test	3

Course Project (25 points)

- Type: Individual or group (max 5 students).
- Goal: Solve a real-world audio- or signal-related problem using methods from the course.
- Grading: 25 points (20 for implementation and results, 5 for presentation/report).

Criterion	Points
Problem Definition & Design	5
Implementation Correctness	10
Experimentation & Results	5
Final Presentation & Report	5

Academic integrity warning

Academic integrity is highly valued by the KSE community and its vast majority of students. We do not tolerate plagiarism and cheating. Very competitive essence of the program calls for absolutely equal

treatment of all students. Even a single case of violation of academic integrity is a serious misdemeanor which may lead to unjust redistribution of grades and, consequently, overall rankings, possible grants, fee reductions and other merit-based awards.

The minimum penalty for a student caught cheating or plagiarizing during any practical task, a presentation or group and final projects will be a zero for this assignment with the consecutive decrease of the final course grade by at least one letter-grade. Repetitive violation of academic integrity may lead to the dismissal of the student from KSE. By enrolling in this course, students affirm that they are familiar with the KSE Code of Academic Integrity.

AI Tools usage policy

Scope

This policy applies to all generative AI tools that produce or modify text, code, images, or other digital content. This includes, but is not limited to:

- *Text generation tools: ChatGPT, Claude, Gemini, Mistral, LLaMA-based models, etc.*
- *Code generation tools: GitHub Copilot, OpenAI Codex, TabNine, etc.*
- *Language assistance tools: Grammarly, DeepL, Google Translate, etc.*
- *Local LLMs or self-hosted AI models.*

Allowed Uses

You may use AI tools for the following purposes, provided they comply with the conditions below:

- *Spell Check & Style Correction. Example: Using Grammarly to correct grammar and improve style in a lab report or paper review.*
- *Translation. Example: Using DeepL or Google Translate to translate sections of a document into another language.*
- *Idea Generation & Pitching. Example: Brainstorming possible approaches for solving a problem or generating research ideas.*
- *Self-Education. Example: Using ChatGPT to explain an unfamiliar algorithm or concept.*
- *Library & Documentation Exploration. Example: Asking AI to summarize PyTorch documentation for a specific class or function.*
- *Code Assistance (Partial). Writing or correcting specific functions or methods (not entire solutions). Examples: Write a function that reads data and converts it to another format. Implement a data collator with padding to the longest sequence. Create an image augmentation function for training.*

Conditions for Use

Full Transparency:

- *Any use of AI tools must be disclosed in your submission.*
- *Include all prompts, responses, and/or screenshots showing your interaction with the AI tool.*
- *If using online tools, include chat links (if available) or full transcripts in an appendix of your report.*

Partial Contribution Only:

- *AI tools should support your work — they must not produce the complete assignment, lab, or report.*
- *You must be able to explain and defend any AI-generated content during oral defense or Q&A*

Examples

Acceptable Uses:

- *"Correct grammar and style in this paragraph of my report."*
- *"Write a function to load images from a directory and resize them."*
- *"Suggest 3 ideas for improving model generalization."*

Unacceptable Uses:

- *"Write the entire peer review for this paper."*
- *"Generate the complete solution code for this assignment task."*

Resources

All materials and resources will be available on course GitHub